PAPER REVIEW

Open Wireless Digital Twin

End-to-End 5G Mobility Emulation with OpenAirInterface and Ray Tracing

Tetsuya Iye et al.

IEEE Access, 2023

System Architecture

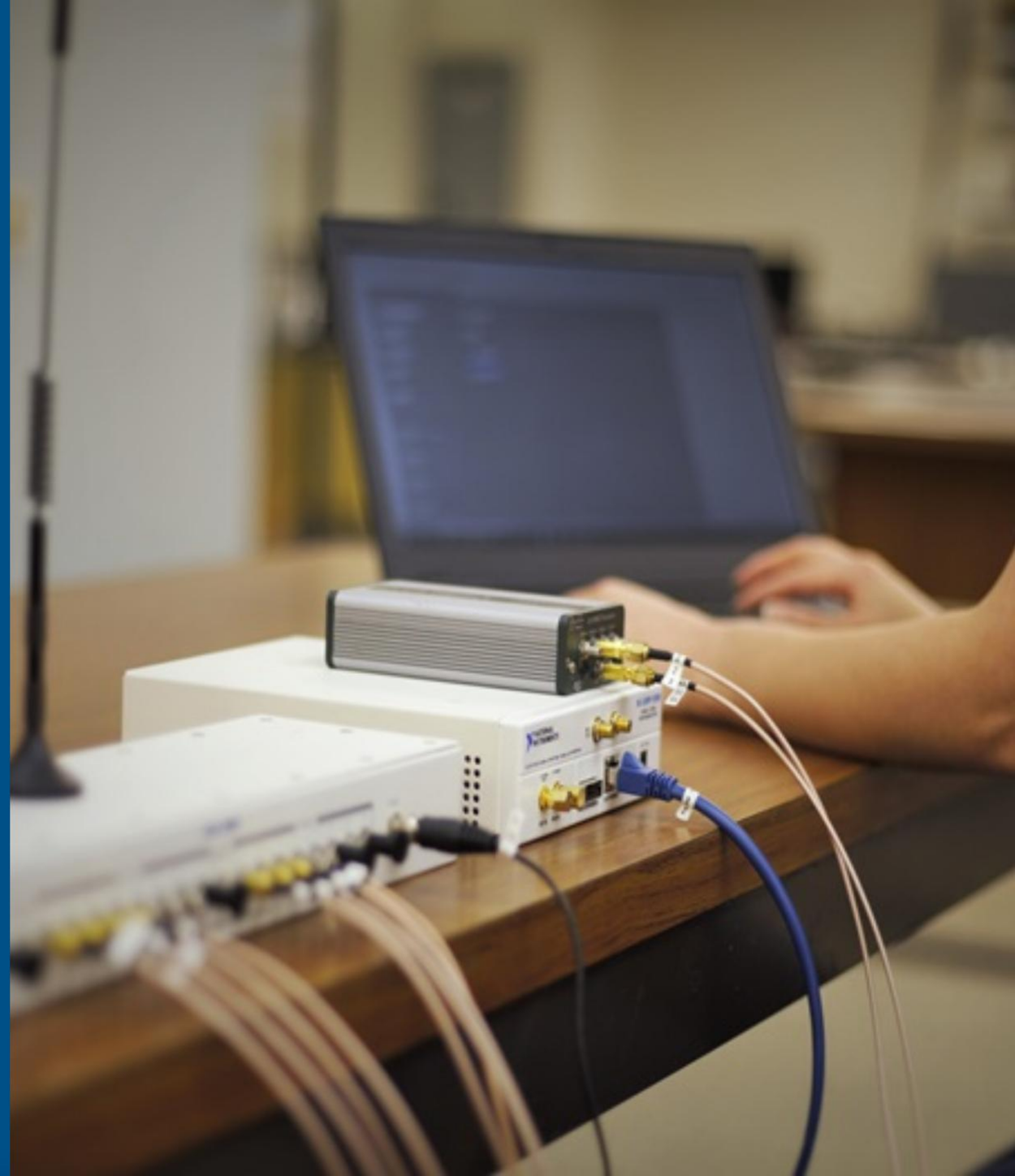
The Problem

Field trials for 5G and Connected Autonomous Vehicles (CAV) are expensive, require licenses, and lack reproducibility. Existing simulators often abstract the physical layer too heavily.

The OWDT Solution

An End-to-End (E2E) digital twin platform integrating:

- **OpenAirInterface (OAI):** 5G Core, RAN, and UE stack.
- **Sionna RT:** High-fidelity ray tracing for propagation.



Emulation Workflow

1. 3D Data

Import urban data (PLATEAU/OSM) into Blender to create material-mapped environments.

2. Mobility

Generate realistic vehicle trajectories and traffic scenarios using SUMO.

3. Ray Tracing

Compute Channel Impulse Response (CIR) via Sionna RT (NVIDIA).

4. Convolution

Apply CIR to baseband IQ signals in OAI gNB in real-time on CPU.

Results & Validation

Scenarios: Shin-Nakano & Shibuya

The system was validated in both residential and dense urban environments. The path gain calculated via ray tracing strongly correlated with the RSRP measured by the OAI UE.

Key Findings

- **High Fidelity:** Reproduced LoS/NLoS transitions and multipath fading effects.
- **Real-Time:** Achieved convolution of up to 28 taps on a standard CPU.
- **O-RAN Ready:** Successfully monitored KPIs (Throughput, BLER) via FlexRIC.
- **Impact:** Democratizes 5G digital twins by removing the need for expensive hardware emulators.

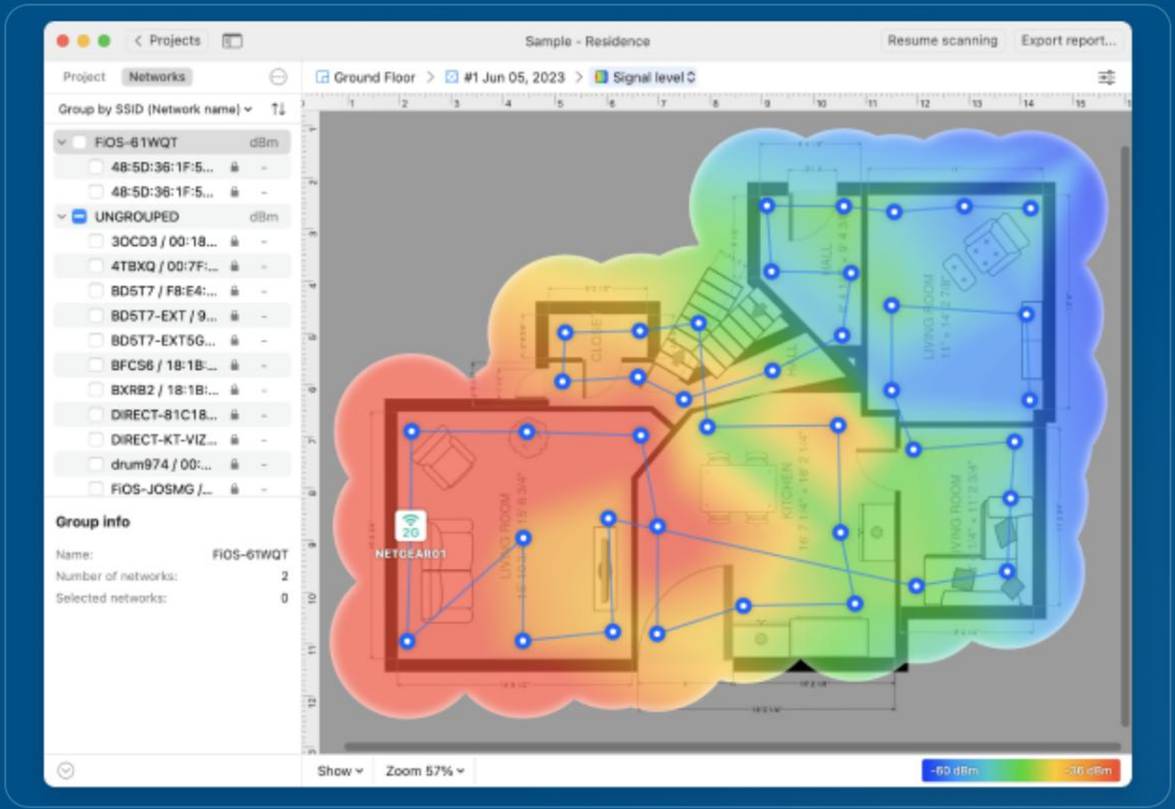


Fig 1. Signal coverage simulation in an urban environment.

Image Sources

https://rktelesystem.com/rktplgroup/wp-content/uploads/2020/09/sdr.jpg

Source: rktelesystem.com

https://www.netspotapp.com/wp-content/uploads/2024/09/NetSpot_Heatmap_signal_level.png

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